PAPER

# Locating and Mitigating Factual Hallucination Circuits in Small Language Models via Activation Patching

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**SUMMARY** Small language models (SLMs) are increasingly deployed in resource-constrained systems, yet they remain prone to factual hallucination, particularly when a misleading context conflicts with parametric knowledge (a context–memory conflict). We present a systematic activation-patching study of this conflict in three SLMs (Gemma-2-2B with 2,000 facts, Llama-3.2-1B and Qwen2.5-1.5B with 600 facts each), each filtered to facts the model demonstrably knows. Restoration effects concentrate in the final quarter of the network (pairwise Spearman correlation of layer profiles 0.81–1.00), dominated by a small set of late-layer attention heads. Bidirectional patching shows that influential heads act in both patching directions (head-level superposition) and separates dual-role heads from the specialist head whose neutral-control effect is minimal. In Gemma-2-2B, knocking out that specialist, L22.H4 (the only head, of the four checked in both bfloat16 and float32, passing the neutral criterion), reduces misleading-context distractor-following by 35% relative (exact McNemar $p$=3.4 × $10^{-23}$) while increasing first-token accuracy by 5.3 percentage points, with MMLU retention 0.992, a 5% perplexity cost, and accuracy under a supportive correct context nearly intact (96.1% vs. 98.2%); in the two replication models, single-head knockouts of each model's own dev-ranked head reduce distractor-following by 30–50% (the specialist characterization itself is Gemma-2-2B-only). Eight of ten random single-head knockouts beat the specialist on both behavioral metrics at comparable capability cost; two carry severe costs (retention 0.70; PPL ×2.3). The work's contribution is thus causal localization, role characterization, and measured trade-offs, not a demonstrated superiority of the selection procedure over random single-head ablation. We report quantified dose–response trade-offs between reduction and capability, with relation-level splits guarding against circular analysis.

***key words:*** *mechanistic interpretability, hallucination, knowledge conflict, activation patching, attention heads*

## 1. Introduction

Language models of 1–2.6B parameters now power on-device assistants [29], retrieval-augmented pipelines [9], and document-processing systems where latency and memory budgets preclude larger models. In such deployments the model must frequently integrate an external context with its parametric knowledge. When the context asserts a false fact (whether through a retrieval error, an adversarial document, or a mistaken user, one way misinformation reaches a model in context [7]), the model faces a *context–memory (CM) conflict* [5], and resolving it incorrectly produces a factual hallucination, which can then cascade into further errors [32].

Manuscript received October 13, 2026.
Manuscript revised October 13, 2026.

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DOI: 10.1587/transinf.E0.D.1

Why a mechanistic account? Deployed systems need interventions that are cheap, targeted, and auditable: knowing *which* components carry the conflict lets a builder trade hallucination against capability with measured confidence. The account is more valuable still if the responsible components transfer across architectures, which is why replication is a first-class concern here.

A growing body of work studies hallucination mechanisms, including knowledge-deficit failures [2] and probes/directions for truthfulness [3], [4]. Head-level analyses of CM conflict exist in text-only models [17], [18] and multimodal settings [19]. What has been missing for this size class is a systematic causal account (which components, located how, trade hallucination against capability when intervened upon, and whether findings replicate across architectures), and, for head-level work generally, a neutral-control separation of head roles and quantified single-head intervention costs.

This paper provides that account with a design that separates localization from validation: heads are selected only on a development split of relations, and all intervention results are reported on held-out relations. Our contributions are:

- **Localization.** A coarse-to-fine activation-patching map of the CM-conflict circuit in Gemma-2-2B, replicated in two other architectures: restoration concentrates in late layers, and attention patches restore more than MLP patches under our protocol (positions differ across corruption regimes), a protocol-bound contrast with the MLP-heavy picture of classical fact recall.
- **Characterization.** Bidirectional patching quantifies per-head asymmetry and separates dual-role heads from the conflict specialist via the neutral control.
- **Quantified trade-offs.** Dose–response curves for knockout and mean-ablation with fairness baselines: patching-guided selection reduces hallucination significantly (exact McNemar $p<10^{-6}$ for single heads in Gemma-2-2B; $p$=0.004/9×$10^{-10}$ in the replications), norm-based selection *increases* it, and capability costs dissociate (MMLU ≈1.00; perplexity +5–20%).

## 2. Related Work

**Hallucination and knowledge conflict.** Surveys taxonomize hallucination in NLG [6] and knowledge conflicts in LLMs [5]. Behavioral studies quantify when context over-

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rides memory [8]; attention-map analysis detects contextual hallucinations [10] and enables quality-improving masking in MT [11]; data-side remedies refine noisy factual resources [12]. Yu et al. trace non-factual hallucination to knowledge deficits in MLPs and selection failures in attention [2]; our corruption regime instead targets *knows-but-overridden* failures. Mechanistic work has also examined conflicts inside parametric memory itself [23] and cautioned that heads appearing to implement factual recall may partly implement generic copying [24], a confound our neutral control tests for at the level of patching effects between conditions (not a pathway isolation).

**Activation patching and circuits.** Causal mediation introduced the method [13]; ROME localized factual associations with causal tracing [1]; best practices are systematized in [14]. Head-level analyses include IOI [15] and copy suppression [16]. Closest to us are two head-level studies of knowledge conflict. JuICE reports that influential heads play superposed dual roles in knowledge conflicts [17]; PH3 identifies memory and context heads by path patching and prunes them to steer conflict resolution in either direction, demonstrating cross-model, cross-relation, and cross-format transfer [18]. Our work differs in what is selected and how the cost is measured: (i) a neutral-control analysis that distinguishes (at the level of patching effects) a head selected by the neutral-effect criterion from dual-role heads, a distinction neither method draws; (ii) single-head knockout of the operationally defined specialist, selected and evaluated under a relation-disjoint dev/held-out split; and (iii) explicit capability trade-off curves (MMLU, perplexity) reported alongside hallucination metrics, which, to our knowledge, neither study reports. Our top-$k$ knockout is structurally close to PH3-style context-head pruning; specialist selection removes 35% of distractor-following with one head while accuracy rises, whereas aggregate pruning trades accuracy for compliance (Sect. 4.3).

**Inference-time intervention.** ITI steers along probe-identified truthful directions [3]; truth directions are linearly represented [4]; hallucination can also be detected from sampling uncertainty [22]. Our interventions are *subtractive* (head knockout/mean-ablation) and selected by causal patching rather than probes, with explicit capability trade-off curves.

**Attention sinks and knowledge editing.** Autoregressive models allocate disproportionate attention to initial tokens ("attention sinks"), which stabilize generation but are usually treated as functionally inert no-ops [20]. Our findings complicate this picture: the strongest conflict specialist dominating the CM-conflict circuit is a BOS-sink head that is causally *active* under misleading context (Sect. 4.2). Separately, knowledge editing locates and rewrites stored facts [1], [21]; we instead locate components involved in *context override* and ablate them at inference time, leaving parameters untouched.

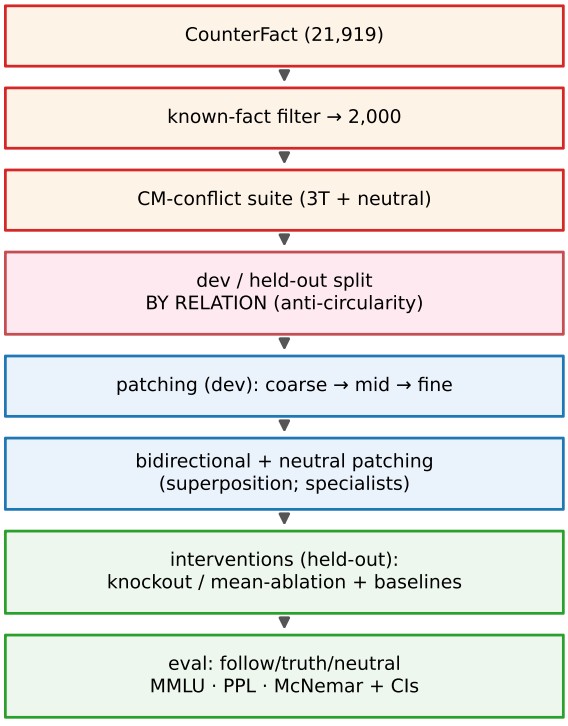



**Fig. 1** Protocol overview. Data construction (top), causal localization on the dev split (middle), and interventions evaluated exclusively on held-out relations (bottom).

## 3. Experimental Setup

Figure 1 summarizes the protocol.

**Models.** Primary: Gemma-2-2B (26 layers, 8 heads) [27]. Cross-model: Llama-3.2-1B (16L/32H) [28], [29], Qwen2.5-1.5B (28L/12H) [30]. All experiments use bfloat16 on a single 24GB GPU. Two separate numerical checks: (i) *framework parity*: our hooking framework vs. reference HuggingFace logits, both in float32 (mean KL $1.3 \times 10^{-5}$, argmax agreement 100%); (ii) *precision robustness* of the reported metrics themselves (Sect. 4.1).

**Data.** From CounterFact [1] (first 12,000 facts of the `wangzn2001/counteract` copy, case order) we retain facts each model demonstrably knows (first-token argmax over the full vocabulary equals the first token of the space-prefixed true target, and the first-token logit gap over the counterfactual distractor $\geq 0.5$), sampling 2,000 facts from the 2,130 that pass both filters (17.8% filter survival; 16.7% of scored facts retained; seed 0), split 807/1,193 dev/held-out *by relation*, so the held-out split shares no relation with the split used for any selection. A CM-conflict suite asserts the distractor in three templates; a neutral control replaces it with a non-answer token.

**Metrics and definitions.** Let $z_t(x)$ and $z_d(x)$ denote the logits, at the answer position of input $x$, of the first tokens of the true target and the distractor, respectively. The *logit*

**Table 1** Comparison with the closest head-level knowledge-conflict studies.

| | JuICE [17] | PH3 [18] | this work |
|---|---|---|---|
| models | larger LMs | 8 LMs | sub-3B SLMs (3 models) |
| head identification | path patching | path patching | activation patching ($z$, suffix-aligned) |
| role characterization | dual roles (superposition) | memory vs. context heads | + neutral control, asymmetry index |
| selection protocol | dev data | dev data, cross-format | dev relations, held-out relation split |
| capability cost measured | not reported | not reported | MMLU + PPL, selected configurations and all random baselines |
| random-head baseline | not reported | not reported | 10 heads, capability-checked |

*difference* is $\mathrm{LD}(x) = z_t(x) - z_d(x)$. Let $c$, $m$, and $n$ denote the clean, misleading, and neutral prompts for a given fact, and let $h$ index a specific attention head. Writing $x \leftarrow y_h$ for the input $x$ with head $h$'s activation replaced by its value from input $y$, the three directed patching effects are:

$$D = \mathrm{LD}(c) - \mathrm{LD}(m), \tag{1}$$

$$A_h = \frac{\mathrm{LD}(m \leftarrow c_h) - \mathrm{LD}(m)}{D}, \tag{2}$$

$$B_h = \frac{\mathrm{LD}(c) - \mathrm{LD}(c \leftarrow m_h)}{D}, \tag{3}$$

$$N_h = \frac{\mathrm{LD}(n \leftarrow c_h) - \mathrm{LD}(n)}{\mathrm{LD}(c) - \mathrm{LD}(n)}. \tag{4}$$

Here $A_h$ is the *restore effect* (patching the clean value into the misleading run; $A$=1 is full restoration), $B_h$ the *context-push effect* (patching the misleading value into the clean run; $B$=1 is full degradation), and $N_h$ the *neutral effect* (patch into the neutral run, normalized by the clean–neutral gap, not $D$ of Eq. (1) used for $A$ and $B$). The *directional asymmetry index* $S = |A - B|/(|A| + |B| + \epsilon)$, with $\epsilon = 10^{-8}$, equals 1 for a head that pushes one direction only and 0 for a balanced dual-role head; low $S$ thus corresponds to strong superposition in the sense of [17], and we use the two framings interchangeably below. When both $|A|$ and $|B|$ are near zero the index is uninformative, and we exclude such heads from superposition-based claims. Effects where the denominator has magnitude $< 10^{-6}$ are set to 0. Head-level values of $A$, $B$, and $N$ are means over per-fact effects; the reported neutral ratio $N/A$ is the ratio of these means (not a mean of per-fact ratios). Hallucination rate is the fraction of misleading-prompt items whose greedy completion equals the distractor (context-following); accuracy is the fraction matching the true target; both are first-token measures, so "accuracy" means first-token accuracy throughout. We use the term "misleading-context distractor-following" for this specific, programmatically verifiable metric, which is a strict subset of open-ended factual hallucination. We use "circuit" to mean the causally localized head set, not a fully traced pathway.

**Protocol.** Stage 1 (coarse): patch residual-stream block outputs layer-by-layer under two corruptions (ROME-style embedding noise and the misleading context), measuring normalized restoration of the true-vs-distractor logit difference. Stage 2 decomposes attention vs. MLP outputs on top layers; Stage 3 patches per-head $z$-vectors; Stage 4 characterizes each top head bidirectionally (Eqs. (2)–(4)); Stage 5 zeroes or mean-ablates selected heads and evaluates hallucination, accuracy, and neutral accuracy on held-out items, with MMLU (2,000-question subset) and WikiText-2 perplexity (1024-token windows) under the same hooks. Statistics: bootstrap 95% CIs from 10,000 resamples at the fact level (clustering the templates of one fact together); two-sided exact binomial McNemar tests against "none", where $b$ counts items the intervention fixes and $c$ items it breaks (per outcome: distractor-following, truth, neutral, or correct-context accuracy). The primary comparison is L22.H4 vs. none; all other configurations are exploratory.

**Reproducibility details.** Misleading templates (distractor $d$, prompt $p$): T0 = "*Earlier you learned that the correct answer is d. p*"; T1 = "*The document states: p d. Now answer correctly: p*"; T2 = "*Hint from the user: I am sure it is d. p*"; the neutral control replaces $d$ with "unknown". The relation-level split uses seed 0: 31 relations, 16 dev (807 facts) and 15 held-out (1,193), zero overlap. Answers are scored by first-token argmax at the final position against the first token of the space-prefixed targets (no autoregressive generation, hence no whitespace/punctuation/alias handling). All retained targets are single words (a verified property, not a filter) and no pair shares a first word, so the comparison is well-defined; known-fact status uses the same level. A greedy 8-token completion check on 300 held-out items (template T0, first 300 in case order, relation-clustered and hence an easier subset with a higher baseline than the intervention tables in Sect. 4.3; the check validates scoring coincidence, not aggregate rates) reproduces first-token accuracy exactly (none 82.3%, L22.H4 83.7%; matching compares the leading word of the continuation, case- and punctuation-stripped, with the target; this is a leading-word check, not open-ended factual accuracy). No true/distractor pair shares a first token: the argmax-plus-gap filter would score such a pair 0 and drop it. Patching positions: the noise regime patches every non-BOS position of the identically-lengthened runs; the misleading regime and head-level patching use the last three tokens, right-aligned across runs. Noise corruption is Gaussian at the embedding layer ($\sigma$ = 3× embedding std, all non-BOS tokens, one draw per case). The top-norm baseline ranks heads by mean absolute $z$-component (averaged over token positions and the head dimension, then over 64 dev bare-fact prompts); the mean-ablation vector is each head's mean $z$ over those prompts pooled across token positions, substituted at all positions during evaluation. Six of the top-10 restore heads pass $|N|$<0.01; among these qualifiers, L22.H4 ranks first on *both* restore effect ($A$, 0.071)

and context-push effect ($B$, 0.143), making it the unambiguous top specialist regardless of which metric is used for selection (the overall restore leader, L25.H4, is excluded by the neutral criterion). spec3 takes the three highest-$A$ qualifiers (L22.H4, L22.H0, L20.H3); rank-based choices, no further threshold. Cross-model profile correlation interpolates each model's per-layer mean-effect profile at 10 evenly spaced depth fractions (0.05–0.95) before Spearman ranking. Head rankings and Stage-4 selection use template T0; random-baseline heads are five draws from the full-head pool followed by five from the late-layer pool (layers 20–25), one continuous RNG sequence (seed 20261014; per-head results in Sect. 4.3), while random-5 uses five separate seeds. Coarse/fine sweeps take the first dev facts in case order (caps 300/150); Stage 4 patches one suffix position at a time (150 dev facts × 3 positions = 450 fact–position records), averaged to head level over records, with bootstrap resampling at the fact level (positions travel with their fact). The replication models apply the same filters with a 600-fact cap sampled from survivor pools of 3,860 (Llama; 28 relations, 14/14 dev/held-out) and 3,151 (Qwen; 28 relations, 14/14). All sweeps take the first dev cases in `case_id` order, a fixed deterministic policy, though case order can correlate with relation blocks. All runs use a single NVIDIA L4 (24 GB). Relation and case IDs per split, head lists for every configuration, MMLU question IDs, per-item records, analysis scripts, and exact package versions are provided in the supplementary material archived with this submission. Models use HuggingFace checkpoints `google/gemma-2-2b`, `meta-llama/Llama-3.2-1B`, `Qwen/Qwen2.5-1.5B` (base, not instruct-tuned) loaded in bfloat16 via TransformerLens 2.18 with `center_unembed=False`, `fold_ln=False`. Head ablation zeroes the head's $z$-vector at *all* token positions. MMLU uses the `cais/mmlu` "all" test split, a fixed 2,000-question subset (seed 0), 0-shot lm-eval format (subject header, lettered choices, "Answer:" suffix), scored by a single forward pass comparing the logits of `A`– `D` at the answer position [25]; WikiText-2 uses the `wikitext-2-raw-v1` test split, 100 non-overlapping windows of 1024 tokens taken sequentially from the joined test text, perplexity from mean NLL over predicted tokens [26]. The parity check between TransformerLens and HuggingFace logits was conducted in float32 (mean KL $1.3\times10^{-5}$); all experiments run in bfloat16. KL quantifies distribution-level divergence between the two framework runs, not the error of our summary metrics directly, so we additionally compared logit differences, patching effects, and head rankings between the two precisions on a fixed dev sample (Sect. 4.1).

## 4. Results

### 4.1 Behavioral landscape and localization

On known facts, bare-prompt accuracy of Gemma-2-2B is 100%. Under misleading context the competition is nearly balanced (47.3% following, 95% CI 46.0–48.5, vs. 46.2% truth) but strongly template-dependent (24.2%/98.4%/19.2%), matching reported persuasion-strength variation [8]; the neutral control stays at 87.4% truth with 1.2% neutral-token copying (Table 2). Template 1 (distractor adjacent to the repeated question) drives following to 98.4%, compatible with a strong recency/copy contribution that this protocol does not isolate. Reference benchmarks (lm-eval-harness): MMLU 49.4%±0.4 (full test set), TruthfulQA MC1/MC2 24.5%/36.3% [31], WikiText-2 PPL 65.3; the 2,000-question intervention subset yields 49.8% (unified capability run); this is subset sampling, not protocol error.

**Table 2** Behavioral landscape on known facts (Gemma-2-2B, $n$=2,000; misleading suite per template).

| condition | context-following | truth |
|---|---|---|
| clean (no context) | — | 100% |
| misleading T0 | 24.2% | 66.4% |
| misleading T1 (adjacent distractor) | 98.4% | 1.1% |
| misleading T2 | 19.2% | 71.0% |
| neutral control | 1.2%† | 87.4% |

†fraction copying the neutral token.

**Table 3** Block-level decomposition (mean restoration effect, top layers).

| block | noise regime | misleading regime |
|---|---|---|
| L22 attention | 0.074 | **0.121** |
| L25 attention | 0.018 | 0.048 |
| L21 MLP | **0.076** | ≤0.01 |
| L20 MLP | 0.058 | ≤0.01 |

**Precision robustness.** Repeating the LD measurements and top-20 head-effect ranking in float32 vs. bfloat16 on the same 200 dev facts: mean |ΔLD| is 0.07–0.08 logits vs. mean |LD| 2.7–8.7 (0.9–3%); head effects rank-correlate at Spearman 0.99 (max per-head difference 0.008), and L22.H4 ranks first by mean restore effect among the compared heads on this 200-fact parity sample, under both precisions (the full-experiment dev ranking in Sect. 4.2 uses 150 facts with per-head averaging over three suffix positions, hence its different leader, L25.H4).

Figure 2 shows the coarse maps: restoration under the misleading regime concentrates in the final layers (peak 0.333 at the last layer); the noise regime peaks earlier (layer 21, 0.227). Table 3 decomposes the top blocks: the misleading-conflict signal is attention-dominated (L22 attention 0.121 vs. ≤0.01 for every MLP block), whereas pure recall restoration under noise mixes MLP and attention almost evenly. In short: under this protocol, attention patching restores more than MLP patching in the conflict regime; because the regimes also differ in patched positions (all non-BOS vs. last three), no matched-position control has been run, so the contrast is protocol-bound rather than mechanistic. Head-level patching then identifies a compact set: {L22.H0 (0.26), L25.H4 (0.24), L20.H3 (0.15), L25.H6 (0.12)} under noise, and {L25.H4 (0.08), L22.H4 (0.07), L22.H0, L20.H3} under the misleading context. Note the partial overlap: three of four noise-top heads reappear in the conflict ranking, and L22.H4 emerges only under conflict.

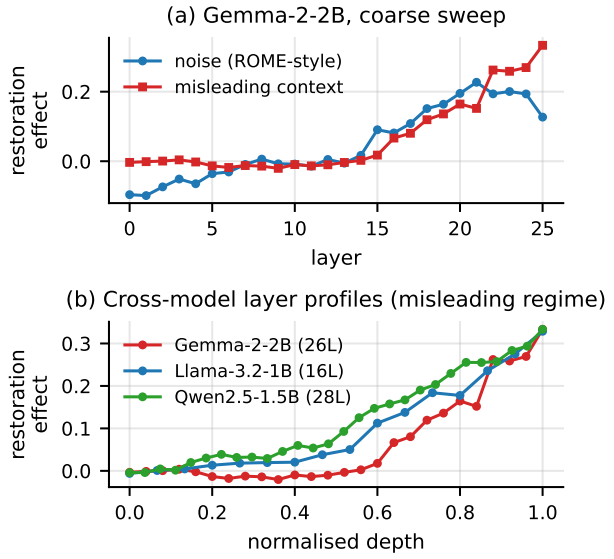



**Fig. 2** Layer profiles of restoration effect. (a) Gemma-2-2B under both corruption regimes; (b) misleading-regime profiles across the three architectures at normalized depth. Late-layer concentration replicates (pairwise Spearman 0.81–1.00).

### 4.2 Superposition and conflict specialists

Bidirectional patching (Fig. 3) shows every influential head conveys the context more strongly than it restores the memory ($B > A$; $S$ 0.21–0.62), consistent with head-level superposition [17]. Under the neutral control, heads can be grouped by the neutral-effect criterion defined below (subject to a normalization caveat); this is an operational grouping, not a statistically demonstrated separation. $N$ divides by the clean–neutral gap (Eq. (4); the absolute value enters only the near-zero guard and magnitude reports), which is far smaller than $D$ (median |gap| 0.75 vs. 6.7 logits; 35% of cases below 0.5; 5.3% with $|\text{gap}|<10^{-6}$ set to $N$=0), so $N/A$ is not a fraction of effect retained; in raw logit units, patching L25.H4's clean value changes the neutral run by only 0.01 logits (0.50 in the misleading run). Near-zero signed means are not mere cancellation: for the *median* case the patch leaves the output logits bit-identical in bfloat16 (median $|\Delta\text{LD}_N|$=0 for every top head, 95% bootstrap CI = $[0, 0]$; float32 medians are 0.002–0.007 logits, so the exact zeros partly reflect rounding), with mean $|\Delta\text{LD}_N|$ of 0.03–0.08 logits driven by a minority of cases. Two distinct thresholds are involved. *Denominator* thresholds (0.5/1/2 logits) change the $N$ estimate itself: cases below them are zeroed, not dropped, leaving 291/186/72 of 450, under which L25.H4's share stays largest ($N$=0.029/0.019/0.006 vs. ≤0.016 for the specialists). *Membership* criteria on $|N|$ instead change which heads qualify: 4/6/9 heads pass $|N|<0.005/0.01/0.02$, and L22.H4 ($N$= − 0.006) is selected under the working criterion $|N|<0.01$ but would be excluded at 0.005. "Specialist" therefore means *low neutral effect under the tested control* (a point estimate; Table 4 gives CIs and the precision check),

**Table 4** Neutral-effect check: signed mean $N$ per head (bfloat16, all top-10, with fact-clustered bootstrap 95% CIs; float32 recomputation for the four heads checked). "Pass" = $|N| < 0.01$ (bf16/fp32).

| head | $N$ (bf16) | 95% CI | $N$ (fp32) | pass |
|---|---|---|---|---|
| L25.H4 | +0.062 | [+0.026, +0.103] | +0.046 | no/no |
| **L22.H4** | −0.006 | [−0.030, +0.016] | +0.006 | **yes/yes** |
| L22.H0 | +0.002 | [−0.031, +0.033] | +0.067 | yes/no |
| L20.H3 | −0.004 | [−0.052, +0.037] | +0.020 | yes/no |
| L23.H7 | −0.012 | [−0.038, +0.012] | — | no/— |
| L23.H4 | −0.013 | [−0.037, +0.011] | — | no/— |
| L20.H7 | −0.009 | [−0.036, +0.018] | — | yes/— |
| L22.H3 | −0.013 | [−0.038, +0.011] | — | no/— |
| L25.H2 | +0.004 | [−0.013, +0.018] | — | yes/— |
| L21.H1 | +0.001 | [−0.028, +0.031] | — | yes/— |

not absence of other functions; its behavioral value is established by the interventions, not the threshold. A float32 recomputation of the neutral effects (150 dev facts) confirms L22.H4 ($N$=0.006, passes; mean $|\Delta\text{LD}_N|$ 0.012 logits) and shows the bf16 exact-zero medians are rounding artifacts (fp32 medians 0.002–0.007 logits, still minuscule); the fp32 values for L22.H0 and L20.H3 (0.067, 0.020), however, no longer pass $|N|<0.01$, so the *additional* members of spec3 are precision-sensitive, while L22.H4 itself is not. Read as each head's *share of the clean–neutral gap*, L25.H4 carries the largest share under every denominator threshold. Among the four heads checked in both precisions (Table 4), L22.H4 is the only one below the working criterion in *both* by point estimate; note, however, that its CI $[-0.030, +0.016]$ extends beyond the ±0.01 criterion, so the classification is operationally defined rather than an equivalence within the threshold. (L25.H2's CI $[-0.013, +0.018]$ is narrower and its point estimate passes in bfloat16, but it was not checked in float32.) spec3 (L22.H4, L22.H0, L20.H3) is a bfloat16-selected exploratory configuration whose additional members are precision-sensitive; its behavioral results stand as reported, without resting on the classification label.

Figure 4 measures, from the answer position of 32 misleading held-out prompts (mean±SD), where each head reads (the first 32 held-out T0 prompts whose distractor matches exactly one token; targets are single words but may tokenize into several tokens, and all occurrences of the matched token are counted). The strongest specialist, L22.H4, concentrates 68 ± 9% of its answer-position attention on ⟨bos⟩ and a further 18 ± 8% on the token asserting the distractor (86% combined); the bfloat16-selected candidates split 22–35% BOS / 10–12% distractor, whereas the dual-role heads L25.H4/L25.H6 read the distractor token only 3%/0.5%. Reading-position attention to the distractor is modest for every head (2–11% averaged over positions following the distractor). We note that BOS precedes the misleading context in the causal decoder, so BOS attention limits *what the head reads* rather than explaining *how* it conveys context information; the mechanism may operate through a query-position output fed by the BOS value (the write happens where the head queries, not at BOS), through attention reallocation, or through downstream residual-stream effects;

**Table 5** Per-head characterization (Gemma-2-2B, top-10 by restore effect $A$; $S$ = directional asymmetry index; $N/A$ = neutral ratio; bfloat16 values; see text for the float32 check).

| head | $A$ | $B$ | $N$ | $S$ | $N/A$ |
|---|---|---|---|---|---|
| L25.H4 | 0.081 | 0.134 | 0.062 | 0.24 | 0.76 |
| L22.H4 | 0.071 | 0.143 | −0.006 | 0.34 | −0.09 |
| L22.H0 | 0.048 | 0.095 | 0.002 | 0.33 | 0.05 |
| L20.H3 | 0.038 | 0.096 | −0.004 | 0.43 | −0.10 |
| L23.H7 | 0.016 | 0.038 | −0.012 | 0.41 | −0.72 |
| L23.H4 | 0.012 | 0.022 | −0.013 | 0.29 | −1.02 |
| L20.H7 | 0.010 | 0.042 | −0.009 | 0.62 | −0.91 |
| L22.H3 | 0.010 | 0.030 | −0.013 | 0.50 | −1.33 |
| L25.H2 | 0.010 | 0.024 | 0.004 | 0.42 | 0.38 |
| L21.H1 | 0.009 | 0.019 | 0.001 | 0.34 | 0.06 |

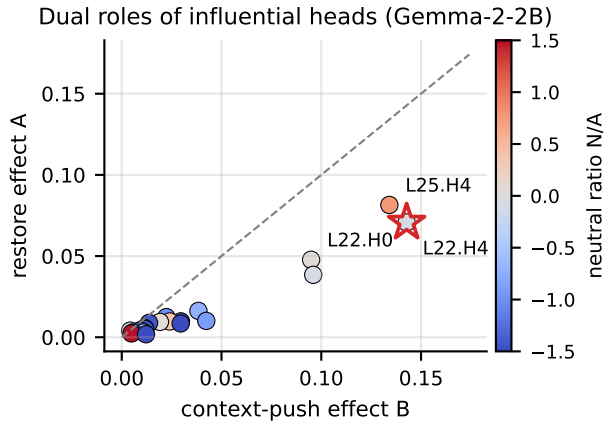



**Fig. 3** Dual roles of the top-20 heads: context-push effect $B$ vs. restore effect $A$, colored by neutral ratio $N/A$. Points near the diagonal are superposed; L22.H4 combines the strongest context conveyance with a minimal effect under the tested neutral control.

separating these hypotheses remains open. Table 5 reports the full per-head characterization for the ten highest-restore heads. Two structure-level observations follow. First, $B > A$ for *every* head tested; none behaves as a pure "memory carrier" under these patching directions; all respond in both directions. Second, the bfloat16-selected spec3 set (L22.H4, L22.H0, L20.H3) combines $N \approx 0$ with mid-range asymmetry ($S$ 0.33–0.43), whereas heads combining large $|N/A|$ with higher asymmetry (L20.H7, L22.H3) sit lower in the restore ranking; though a large $|N/A|$ can also reflect a small $A$, so this reads as an association, not a causal role claim.

### 4.3 Interventions and quantified trade-offs

Table 6 and Fig. 5 report the held-out interventions. Knocking out the dev-ranked top-$k$ heads reduces context-following monotonically (19.3%→1.6% at $k$=20), with overall accuracy loss as $k$ grows (truth 74.4% at none → 66.8% at $k$=1 → 19.4% at $k$=20, with a small non-monotonic bump at $k$=2). The Stage-4 taxonomy predicts that removing a conflict specialist suppresses the misleading pull at limited collateral cost: knocking out the single head L22.H4 reduces hallucination to 12.6% (two-sided exact McNemar

**Table 6** Held-out interventions (Gemma-2-2B, misleading template T0; all values %). $n$=1,193 misleading + $n$=1,193 neutral items per config; follow rates carry fact-clustered bootstrap 95% CIs. Baselines differ from Table 2 because this table covers held-out facts only.

| config | follow↓ (95% CI) | truth↑ | neutral |
|---|---|---|---|
| none | 19.3 (17.0–21.5) | 74.4 | 93.5 |
| top-1 | 15.8 (13.8–17.9) | 66.8 | 82.6 |
| top-2 | 11.7 (9.8–13.5) | 69.9 | 80.1 |
| top-5 | 8.5 (7.0–10.1) | 44.7 | 50.9 |
| top-10 | 4.5 (3.4–5.8) | 32.0 | 39.6 |
| top-20 | 1.6 (0.9–2.3) | 19.4 | 25.9 |
| **spec1 (L22.H4 only)** | **12.6 (10.7–14.5)** | **79.7** | **92.0** |
| spec3 | 10.9 (9.1–12.7) | 63.3 | 68.1 |
| mean-abl. top-2 | 11.2 (9.5–13.1) | 70.3 | 79.4 |
| mean-abl. top-5 | 8.8 (7.2–10.5) | 46.4 | 53.8 |
| mean-abl. spec3 | 11.4 (9.6–13.2) | 63.6 | 70.2 |
| topnorm-5 | 33.0 (30.4–35.8) | 59.1 | 85.9 |
| random-5 (5 seeds) | 12.6–22.2 | 44.8–78.8 | — |
| random-1 (5 draws) | 5.2–12.2 | 81.9–88.3 | — |
| late-rand-1 (5 draws) | 9.3–10.3 | 83.2–84.3 | — |

$b$=81, $c$=1, $p$=3.4 × $10^{-23}$) while *increasing* accuracy to 79.7% (+5.3 points; the gain is itself significant: $b$=81, $c$=18, $p$=1.0×$10^{-10}$) and leaving neutral accuracy at 92.0% (baseline 93.5%; the 1.5-point decrease is small but significant, $b$=5, $c$=23, $p$=9.1 × $10^{-4}$). Fact-clustered 95% CIs for L22.H4: follow 10.7–14.5%, truth 77.5–82.0%, neutral 90.4–93.5%. By contrast, ablating the five highest-norm heads *increases* hallucination to 33.0% ($b$=9, $c$=173): these are early/mid-layer high-$|z|$ heads (L2–L16), not the late-layer heads of Table 5, and their removal tips the final balance toward the context. The random *five*-head ablation (random-5) is erratic across its five seeds (follow 12.6–22.2%, truth 44.8–78.8%, inconsistent sign and magnitude). For L22.H4, the paired fact-clustered 95% CI of the difference is [+5.3, +8.1] points of following removed and [+3.7, +6.9] points of truth gained; per-relation, following falls in 11 of 15 held-out relations (median 4.2 points, range 0–16.7); broad but not relation-uniform; all CIs are conditional on this fixed relation split.

**How special is the specialist?** A uniformly random *single*-head knockout reduces following for every seed tested (5.2–12.2%; truth 81.9–88.3%), and layer-matched late-layer random heads behave the same (9.3–10.3%): reducing distractor-following by one-head $z$-knockout is *not* unique to the specialist; all ten randomly drawn heads tested weaken it. With capability measured for all ten random heads (Table 7),

the picture is sharper: eight of the ten are capability-clean (MMLU retention 0.996–1.01, PPL ratio 0.99–1.05) *and* behaviorally better than L22.H4; two carry large metric-specific costs (L17.H3: MMLU retention 0.70 with slightly *improved* PPL; L5.H4: PPL ratio 2.31 at retention 0.95); damage in one metric need not imply damage in the other. Unprincipled single-head selection is therefore high-variance: most draws are behaviorally effective and capability-preserving, a minority carry large metric-specific costs. In this experiment L22.H4 falls in the low-cost region

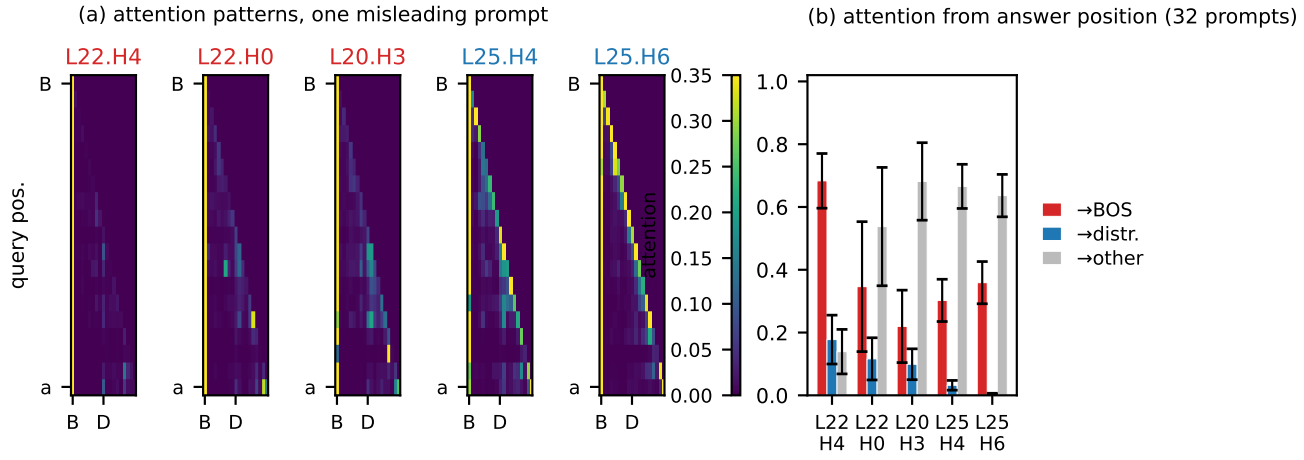



**Fig. 4** Attention of the five top heads. (a) Patterns on a representative misleading prompt (true: French, distractor: English); axis labels: B = ⟨bos⟩, D = distractor token, a = answer query. (b) Attention from the answer position to BOS / the distractor token / other positions, mean±SD over 32 misleading held-out prompts. L22.H4 (red) combines the strongest BOS concentration with direct distractor reading; dual-role heads (blue) read the distractor negligibly.

**Table 7** Random single-head knockouts, per head (ten draws from one seeded RNG, seed 20261014; late = layers 20–25). Follow/truth in %; retention and PPL ratio from the unified capability run.

| head | follow↓ | truth↑ | MMLU ret. | PPL ratio |
|---|---|---|---|---|
| L23.H2 | 9.8 | 83.9 | 0.996 | 0.99 |
| L17.H3 | 5.2 | 88.3 | 0.695 | 0.88 |
| L5.H4 | 10.6 | 81.9 | 0.947 | 2.31 |
| L21.H5 | 12.2 | 82.1 | 0.998 | 1.01 |
| L24.H4 | 9.6 | 83.2 | 1.010 | 1.03 |
| L21.H1 | 10.0 | 83.2 | 0.998 | 1.00 |
| L23.H6 | 9.7 | 83.8 | 1.008 | 1.02 |
| L24.H5 | 9.3 | 84.3 | 1.002 | 0.99 |
| L24.H0 | 9.9 | 83.8 | 1.003 | 1.05 |
| L22.H5 | 10.3 | 83.2 | 1.005 | 1.05 |

(retention 0.992, PPL ×1.05); with one selected head we cannot show that the selection procedure itself controls this variance; the dev-side patching evidence identifies conflict-relevant heads, but whether such screening systematically yields safer heads than random draws is not established here.

**Why do the sampled late-layer knockouts help?** A hypothesis consistent with the results: the dataset is filtered to facts the model answers with 100% bare-prompt accuracy, so parametric memory is intact by construction. Perturbing the sampled heads in the late layers (where our localization places the context-conveyance pathway) may weaken the in-context signal relative to that intact prior, tilting the competition back toward the memorized answer. This is consistent with the uniform behavioral improvement across the ten sampled random heads and with the contrast to topnorm-5, whose early/mid-layer heads (L2–L16) damage the recall side instead and *increase* following to 33.0%; it remains a hypothesis, however, as the pathway itself has not been isolated.

Capability costs are dissociated (Table 8): MMLU retention remains 0.99–1.01 for the patching-guided configurations evaluated there (the random-1 summary row is excluded), and a paired per-question comparison for L22.H4 detects no change on this subset (paired difference −0.40 points, 95% CI [−1.10, +0.30]; $b$=22 gained, $c$=30 lost; exact $p$=0.33); no detected change, not demonstrated equivalence; whereas perplexity rises 5–20%. Among the patching-guided configurations, the single-head specialist is the cheapest: a 35% relative hallucination reduction, with an accuracy *gain*, at MMLU retention 0.992 and a 5% perplexity cost.

**Correct-context control.** A correct-context suite (T0 asserting the *true* target, same held-out facts) tests damage under context that is actually right. Baseline is near ceiling (98.2% truth); L22.H4 leaves 96.1% (small but significant, $b$=0, $c$=25, $p=6\times10^{-8}$; far below its 5.3-point misleading-context gain), whereas spec3 (81.6%) and top-5 (60.2%) destroy it. Because this context concerns already-known facts, the result bounds damage under consistent context rather than measuring context *use* per se; deployment beyond this counterfactual regime (e.g., general RAG reliability) is out of scope.

**Template robustness.** Re-evaluating the none, L22.H4, and top-5 conditions on the two remaining held-out templates tests whether the specialist's effect depends on how the context asserts the distractor. On T2 (user-hint template) L22.H4 again reduces following (16.6%→11.1%) while raising accuracy by 3.3 points, replicating T0. On T1 (the extreme adjacent-distractor template whose baseline behavior is nearly pure context-following at 98.4%, consistent with, though not isolated by, a recency-copying route), L22.H4 reduces following from 98.4% to 89.4% (9.0 percentage points; 9% relative) and raises accuracy from 0.8% (held-out baseline; the all-facts value in Table 2 is 1.1%) to 8.7%, while top-5 reduces following from 98.4% to 58.3%

**Table 8** Capability under intervention, all from one unified run (fixed 2,000-question MMLU subset with per-question records; PPL over 100 non-overlapping 1024-token WikiText-2 windows). The random-1 row spans all ten heads tested (Sect. 4.3).

| config | MMLU | retention | PPL ratio |
|---|---|---|---|
| none | 49.8% | 1.000 | 1.00 |
| top-2 | 49.7% | 0.998 | 1.11 |
| top-5 | 50.2% | 1.007 | 1.20 |
| **spec1 (L22.H4 only)** | **49.4%** | **0.992** | **1.05** |
| spec3 | 49.8% | 1.000 | 1.16 |
| mean-abl. top-5 | 50.2% | 1.007 | 1.20 |
| random-1 (10 heads) | 34.6–50.3% | 0.70–1.01 | 0.88–2.31 |

**Table 9** Cross-model replication summary (misleading regime, held-out).

| | Gemma-2-2B | Llama | Qwen |
|---|---|---|---|
| layers | 26 | 16 | 28 |
| top coarse layers | 20–25 | 13–15 | 22–27 |
| dev/held-out | 807/1,193 | 144/456 | 184/416 |
| sweep facts | 300/150 | 144/144 | 184/150 |
| head | L22.H4* | L15.H7 | L27.H2 |
| follow (none) | 19.3% | 6.6% | 14.9% |
| follow (selected head) | 12.6% | 4.6% | 7.5% |
| Δ truth (pts) | +5.3* | −1.3 | +0.8 |
| MMLU retention | 0.992* | 1.01 | 1.00 |

*Gemma-2-2B uses the Stage-4-selected specialist (L22.H4, not the dev-ranking top-1); the other models use their own dev-ranking top-1; the Gemma-2-2B retention shown is L22.H4's for like-for-like comparison. All coarse/fine selection runs use the dev split only (capped at 300/150 facts; see Sect. 3). Llama = Llama-3.2-1B, Qwen = Qwen2.5-1.5B.

(40.1 percentage points; 41% relative). The measured finding is that single-head ablation effect depends on the prompt format, and substantial residual following remains on T1; whether a copying pathway explains this requires experiments that isolate it (the templates differ in more than distractor position).

### 4.4 Cross-model replication

Replicating the pipeline on Llama-3.2-1B and Qwen2.5-1.5B (600 known facts each, split 144/456 and 184/416 dev/held-out by relation; coarse and fine sweeps use the dev split only, capped at 300/150 facts) confirms the core structural and interventional findings, though the specialist taxonomy (Stage 4) was run only for Gemma-2-2B (Table 9). First, late-layer concentration holds in all three architectures: the most influential coarse layers are 20–25 of 26 (Gemma-2-2B), 13–15 of 16 (Llama), and 22–27 of 28 (Qwen), and the normalized-depth profiles rank-correlate at 0.81 (Gemma-2-2B vs. each other model) and 1.00 (Llama vs. Qwen). Second, baseline following varies across models (6.6% and 14.9%, both below Gemma-2-2B's 19.3%; the splits are relation-disjoint but not item-identical; each model's filter retains different facts (456/416 vs. 1,193 held-out); so differences mix persuadability with data composition). Third, and most importantly, the single-head knockout from each model's own dev ranking reduces distractor-following by 30% (Llama: 6.6%→4.6%, 95% CI 2.9–6.6, exact McNemar $b$=9, $c$=0, $p$=0.004) and 50% (Qwen: 14.9%→7.5%, CI 5.0–10.1, $b$=31, $c$=0, $p=9\times10^{-10}$) with small accuracy changes (−1.3 and +0.8 points) and MMLU retention 1.01/1.00; the single-head effect replicates across architectures; whether dev-ranked selection outperforms random single heads (as it does not, on average, in Gemma-2-2B) is not established for the replication models, where no random baseline was run. Per-head restoration effects are smaller in the 1–1.5B models (0.04–0.06 vs. 0.08); this is consistent with a more distributed conflict circuit at smaller scale, though the data do not separate that reading from generally weaker per-head effects.

## 5. Discussion

**Mechanism.** Under our patching protocol, restoration concentrates in late-layer attention outputs at the answer position, and the influential heads convey the misleading context at least as much as they carry memory ($B > A$ throughout). "Restoration" by patching therefore partly works by displacing a head's context conveyance rather than injecting memory; the neutral-control analysis makes this distinction measurable at the level of patching effects between conditions; it does not by itself identify the underlying pathway.

**Attention-sink specialist.** The most surprising finding is the identity of the strongest conflict specialist: L22.H4, whose answer-position attention is dominated by the beginning-of-sequence token (68 ± 9%; the bfloat16-selected candidates show only 22–35%). In the streaming-efficiency literature such sinks are treated as stabilizing but semantically inert no-ops [20]. Our results show that a sink head can be causally significant here: L22.H4 conveys the strongest context push ($B$=0.143), and its removal improves accuracy at MMLU retention 0.992 (PPL ×1.05). But $N\approx0$ under one neutral-patching control only shows that this intervention had minimal effect in the tested condition, not that the head has no other function. The mechanism by which a BOS-attending head influences the conflict remains an open question. The BOS value vector itself cannot encode the misleading context (which follows it in the causal mask); plausible routes are a query-position output bias fed by the BOS value (the write happens where the head queries, not at BOS), attention redistribution, or downstream effects; and although Figure 4(b) aggregates 32 prompts, mechanism-level claims need larger samples. What the results do establish is that sink-focused heads should not be excluded from behavioral circuit searches.

**Design implication.** The dissociation between MMLU (no detected change on the tested subset) and perplexity (up 5–20%) suggests these heads support contextual integration that matters for language-modeling loss more than for benchmark accuracy; and the norm-selection failure warns that cheap importance proxies can worsen the targeted behavior.

**Limitations and future work.** (i) English-only, greedy completion-style evaluation; (ii) hallucination is operational-

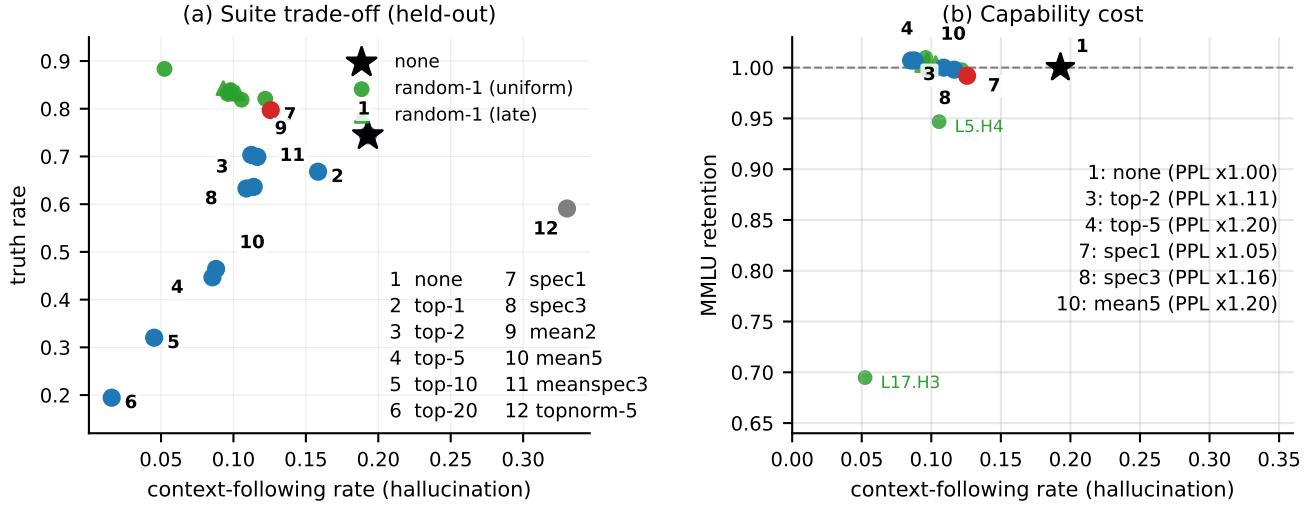



**Fig. 5** Trade-offs; color coding is identical in both panels: blue = patching-guided configurations, red = the specialist L22.H4 (7), gray = norm-selected (12), green = random single-head knockouts (circles: uniform draws; triangles: late-layer draws), shown for comparison but excluded from the patching-guided ranking. (a) Suite space (numbered configurations keyed at bottom right): the specialist attains the highest accuracy among the ranked configurations while removing a third of distractor-following (top-2 removes more following, 11.7 vs. 12.6%, but at 9.8 points lower accuracy, so neither Pareto-dominates). (b) Capability space: hallucination vs. MMLU retention; keys list PPL ratios.

ized as counterfactual override, not open-ended factuality; (iii) head selection is causal but the circuit description is coarse; path-level analysis is the natural next step; (iv) Gemma-2-2B is not the newest generation (chosen for tooling maturity and literature comparability); newer families and instruction-tuned variants are future work; (v) capability is proxied by MMLU/PPL subsets; (vi) the Stage-4 specialist taxonomy ran only for Gemma-2-2B, and entity-popularity effects remain open; (vii) noise and misleading regimes are patched at different positions, so a matched-position control is needed before the attention-vs-MLP contrast can be read mechanistically; (viii) all results use a single fixed relation split with CIs conditional on it; re-splitting and relation-level hierarchical bootstrap would test generality; (ix) the selection procedure was evaluated on one split with one selected head; multi-split comparisons of selection vs random draw are needed to determine whether causal screening systematically reduces capability risk.

## 6. Conclusion

We mapped and exploited the CM-conflict circuit of small language models. Late-layer localization and single-head intervention effectiveness replicate across all three architectures (following falls 30–50% under each model's dev-ranked head); the specialist characterization itself is established only in Gemma-2-2B, where L22.H4, the sole head (of the four checked in both precisions) passing the neutral criterion, can be knocked out individually, reducing distractor-following by 35% on the standard template (9% on the extreme adjacent-distractor template), first-token accuracy rises for L22.H4 (+5.3 points; the replication models' top heads: −1.3 to +0.8) at MMLU retention near 1.00, while norm-based selection backfires. Most random single heads match or beat the specialist behaviorally at similar cost; a minority degrade capability severely. The contribution is the causal characterization and measured trade-offs, not a selection procedure proven superior to random ablation. All selection was performed on dev relations with results reported on held-out relations (CIs are conditional on this fixed split), and the core findings (late-layer concentration, single-head intervention effectiveness) replicated across three architectures. Split IDs, head lists, per-item records, analysis scripts, and environment details accompany this submission as supplementary material; the full experimental codebase will be released publicly upon publication.

## Acknowledgment

This research project is supported by Science Research and Innovation Fund. Agreement No. FF68/KKC/006.

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